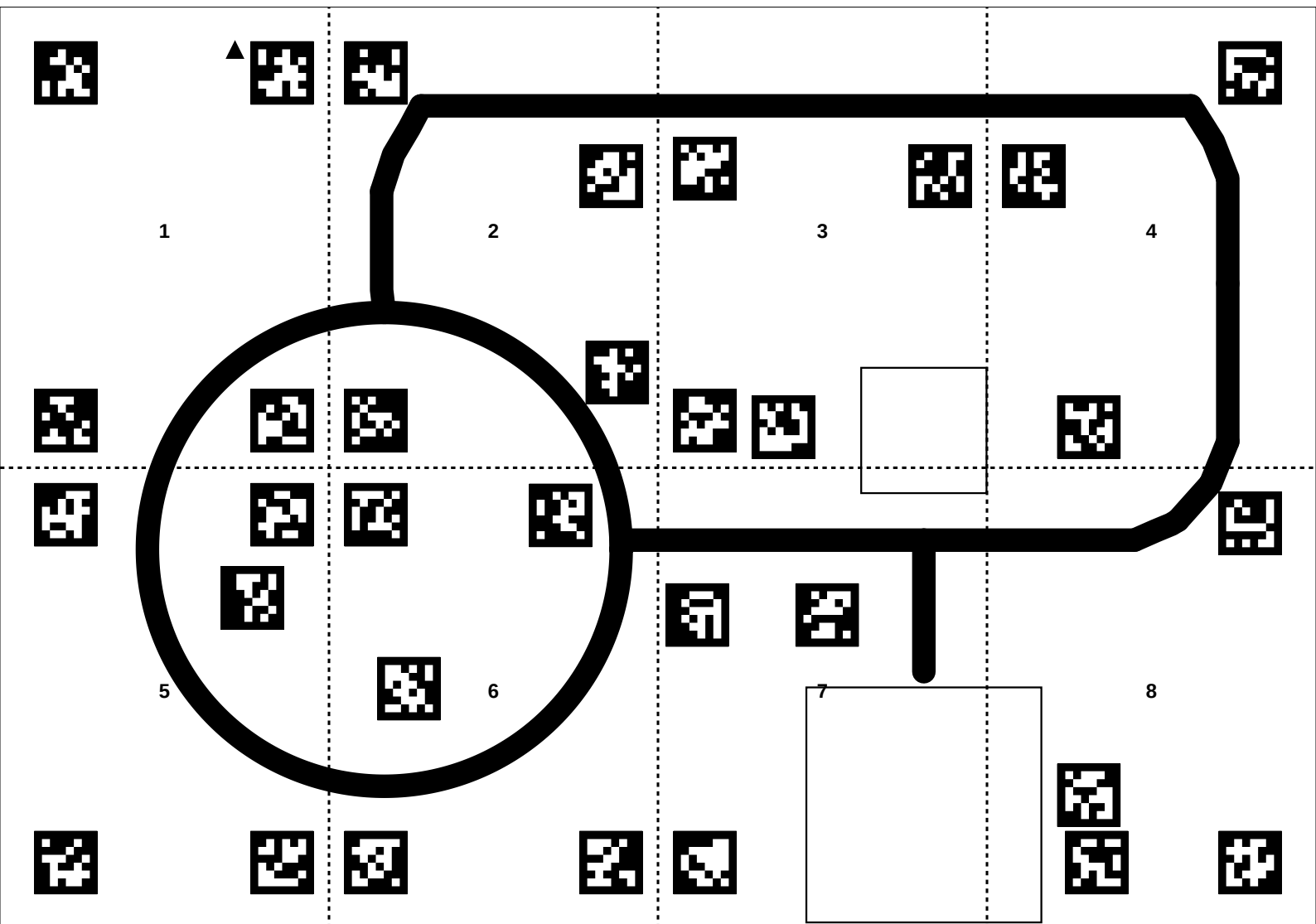


Task-1 Map — Overview

Full-map overview — tile layout reference



Assemble order

Top row (north): tiles 1 – 4, left to right.

Bottom row (south): tiles 5 – 8, left to right.

Align the corner crosses on adjacent tiles and tape on the back.

Task-1 Map — Print & Assemble Guide

1. Print the 8 map tiles (PDF pages 4–11)

Print PDF pages 4–11 on A4 at 100% / Actual Size. Do NOT use “fit to page”.

Verify the scale bar (X 80 mm / Y 40 mm) with a ruler before assembling.

If the corner crosses are clipped, enable borderless printing or reduce printer margins.

2. Assemble (4 columns x 2 rows) — see the overview on PDF page 1

Top row (north) = tiles 1–4 left to right; bottom row (south) = tiles 5–8 left to right.

Align the corner crosses on adjacent tiles; tape on the back.

3. Orientation

The N arrow points map-north (up). Tags are printed with ID upright facing north.

4. AprilTag landmarks (the printed squares)

32 AprilTags (family 36h11, 40 mm) are printed onto the map as absolute-position landmarks.

Each is labelled “ID n (N ↑)” (yaw = 0 = facing map-north).

Source: AprilTag is an open-source project by the University of Michigan APRIL lab.

Official repository: <https://github.com/AprilRobotics/apriltag>

Web search: “AprilTag”, “AprilTag 36h11”, “AprilTag pose estimation”, “OpenCV aruco AprilTag”.

5. Scale

Map 840 x 588 mm = original 1000 x 700 mm Task-1 map scaled 0.84.

Route black line 15 mm; AprilTags 40 mm (physical sizes unchanged).

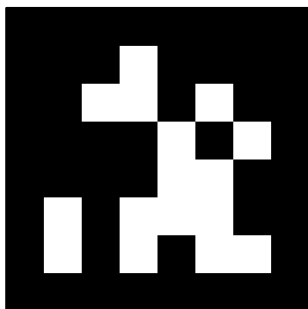
AprilTag Coordinates

X and Y are in millimetres from the map's SOUTH-WEST corner (X east, Y north); NE corner = (840, 588). cm = mm ÷ 10.

ID	X (mm)	Y (mm)	ID	X (mm)	Y (mm)
0	42.0	546.0	21	357.9	263.6
1	180.0	546.0	22	240.0	42.0
2	42.0	324.0	23	390.0	42.0
3	180.0	324.0	24	445.0	200.0
4	240.0	546.0	26	450.0	42.0
6	240.0	324.0	27	528.0	200.0
7	394.0	354.8	28	695.0	85.0
10	450.0	324.0	29	798.0	258.5
11	500.0	320.0	30	700.0	42.0
13	798.0	546.0	31	798.0	42.0
14	695.0	320.0	5	390.0	480.0
16	42.0	264.0	8	450.0	485.0
17	180.0	264.0	9	600.0	480.0
18	42.0	42.0	12	660.0	480.0
19	180.0	42.0	15	161.0	211.0
20	240.0	264.0	25	261.0	153.0

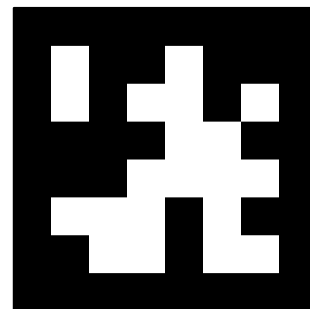
Same values as scratch/landmarks/task1-tag-map.json (metres = mm ÷ 1000).

⊕ tile 1/8



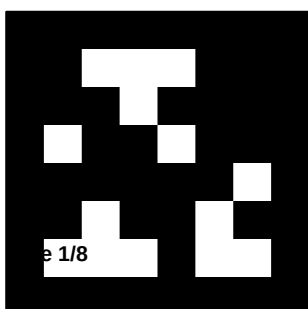
ID 0 (N ↑)

⊕ tile 1/8



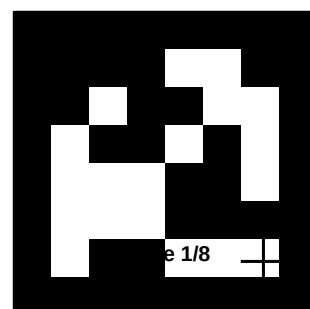
ID 1 (N ↑)

ID 2 (N ↑)

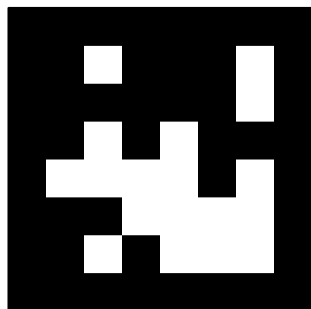


⊕ tile 1/8

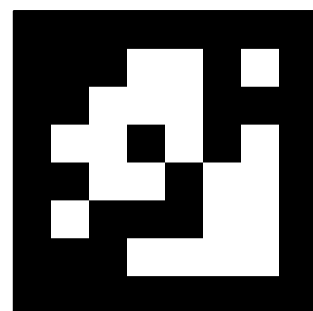
ID 3 (N ↑)



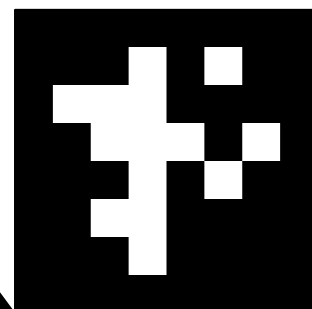
⊕ tile 1/8



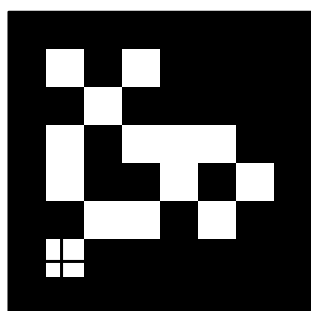
ID 4 (N ↑)



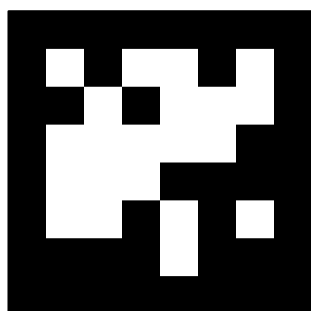
ID 5 (N ↑)



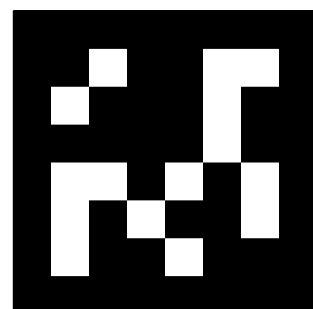
ID 7 (N ↑)



ID 6 (N ↑)

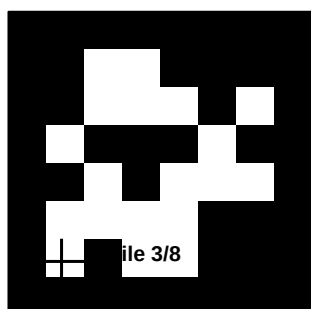


ID 8 (N ↑)

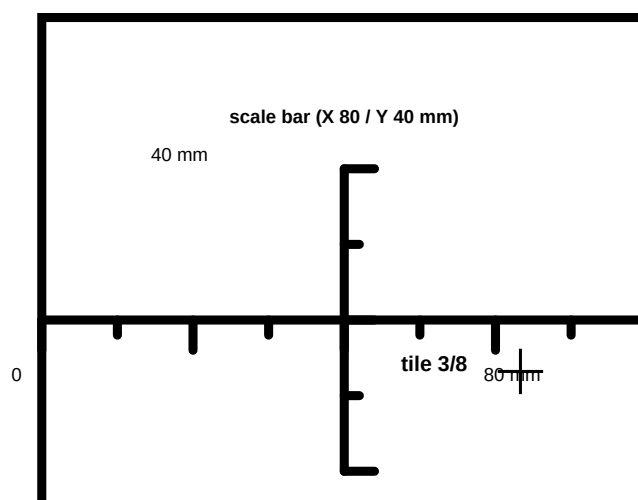
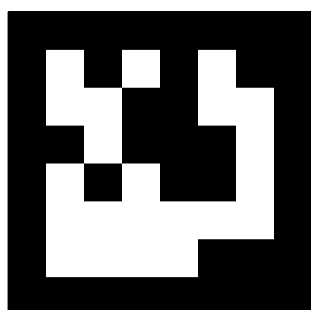


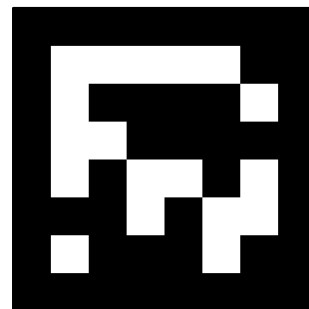
ID 9 (N ↑)

ID 10 (N ↑)

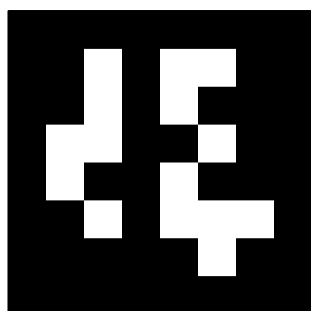


ID 11 (N ↑)



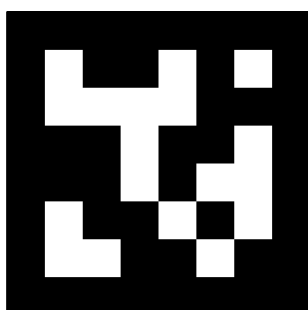


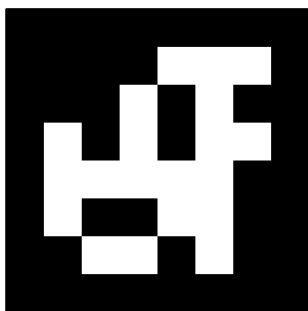
ID 13 (N ↑)



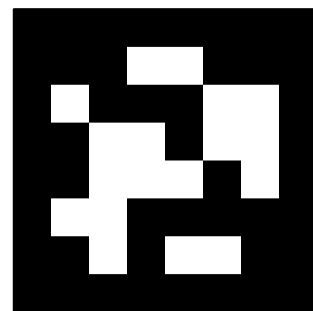
ID 12 (N ↑)

ID 14 (N ↑)

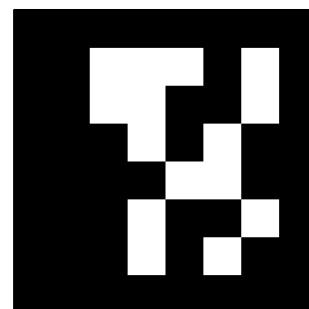




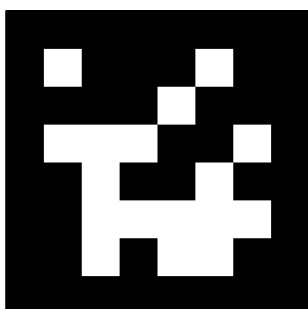
ID 16 (N ↑)



ID 15 (N ↑) ID 17 (N ↑)

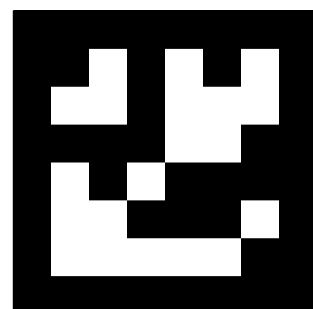


ID 18 (N ↑)



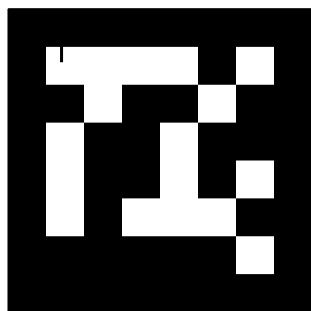
tile 5/8

ID 19 (N ↑)

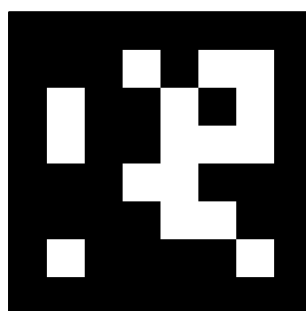


tile 5/8

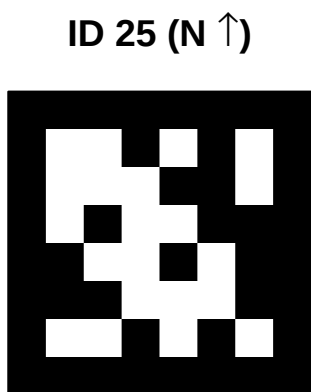




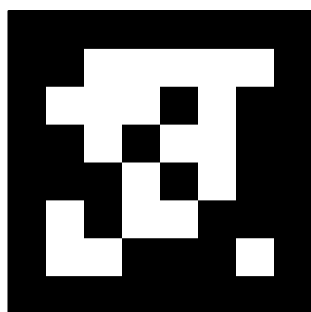
ID 20 (N ↑)



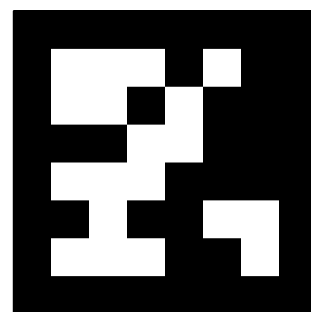
ID 21 (N ↑)



ID 25 (N ↑)

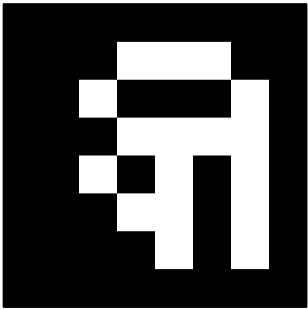


ID 22 (N ↑)

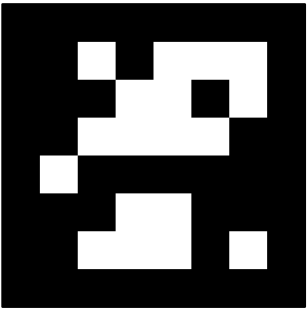


ID 23 (N ↑)

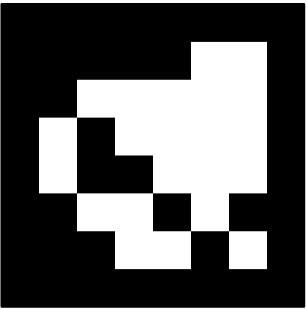
ID 24 (N ↑)



ID 27 (N ↑)



ID 26 (N ↑)



DEPARTURE AREA

START

